COHOMOLOGICAL DESCENT FOR THE COMPARISON OF THE PH-TOPOLOGY WITH THE SMALL ETALE TOPOS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction to a single torsion sheaf, the rewriting of a_{X,*}a_X^{-1}F, and the degeneration of the relative Leray spectral sequence with the two vanishing inputs.

Editorial note. The label lemma-cohomological-descent-etale-ph names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the six cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Torsion complexes have torsion representatives. Cohomology on Sites, the lemma representing a complex with torsion cohomology sheaves by a bounded below complex of torsion abelian sheaves. Supplies the representative used in the reduction step.
- The ph-description of the pullback and the ph V-C statement. Same chapter, the description of the pullback in the ph topology and the V-C statement giving the vanishing of the higher pushforwards of epsilon_X there. Supplies the identities a_{X,*}a_X^{-1}F = F and the vanishing of the higher pushforwards of the two factors.
- Leray's acyclicity lemma and the relative Leray spectral sequence. Derived Categories, the acyclicity lemma, and Cohomology on Sites, the relative Leray spectral sequence for a_X = epsilon_X o pi_X. Supplies the reduction to a single sheaf and the vanishing of the higher direct images of a_X.

Source reference. Original lemma 015E: sources/stacks-chow/tags.txt, source lines 1505–1505. Prerequisite pointer invoked at proof line(s) 22460; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DE4: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 22389–22389. Prerequisite pointer invoked at proof line(s) 22472; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 0DDI: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21308–21308. Prerequisite pointer invoked at proof line(s) 22475; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DDW: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 22175–22175. Prerequisite pointer invoked at proof line(s) 22466; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0734: sources/stacks-chow/tags.txt, source lines 9048–9048. Prerequisite pointer invoked at proof line(s) 22471; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DD7: sources/stacks-chow/tags.txt, source lines 16757–16757. Prerequisite pointer invoked at proof line(s) 22455; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme. For $K \in D^+(X_{étale})$ with torsion cohomology sheaves the map*

$$K \longrightarrow Ra_{X,*}a_X^{-1}K$$

is an isomorphism with $a_X : Sh((Sch/X)_{ph}) \to Sh(X_{étale})$ as above.

Proof. We first reduce the statement to the case where K is given by a single abelian sheaf. Namely, represent K by a bounded below complex $\mathcal{F}^\bullet$ of torsion abelian sheaves. This is possible by Cohomology on Sites, Lemma 0DD7. By the case of a sheaf we see that $\mathcal{F}^n = a_{X,*}a_X^{-1}\mathcal{F}^n$ and that the sheaves $R^q a_{X,*}a_X^{-1}\mathcal{F}^n$ are zero for $q > 0$. By Leray's acyclicity lemma (Derived Categories, Lemma 015E) applied to $a_X^{-1}\mathcal{F}^\bullet$ and the functor $a_{X,*}$ we conclude. From now on assume $K = \mathcal{F}$ where $\mathcal{F}$ is a torsion abelian sheaf.

By Lemma 0DDW we have $a_{X,*}a_X^{-1}\mathcal{F} = \mathcal{F}$. Thus it suffices to show that $R^q a_{X,*}a_X^{-1}\mathcal{F} = 0$ for $q > 0$. For this we can use $a_X = \epsilon_X \circ \pi_X$ and the Leray spectral sequence (Cohomology on Sites, Lemma 0734). By Lemma 0DE4 we have $R^i\epsilon_{X,*}(a_X^{-1}\mathcal{F}) = 0$ for $i > 0$ and $\epsilon_{X,*}a_X^{-1}\mathcal{F} = \pi_X^{-1}\mathcal{F}$. By Lemma 0DDI we have $R^j\pi_{X,*}(\pi_X^{-1}\mathcal{F}) = 0$ for $j > 0$. This concludes the proof. □